



Immunoglobulin administration for the treatment of CIDP: IVIG or SCIG?

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ABSTRACT

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an acquired neurological disorder characterized clinically by weakness and impaired sensory function evolving over 2 months or more, loss or significant decrease in deep tendon reflexes, and by electrophysiological evidence of peripheral nerve demyelination. Expedient diagnosis and treatment of CIDP early in the disease course is critical such that irreversible disability can be avoided. Intravenous immunoglobulin (IVIG) is one first-line and maintenance therapy option for CIDP. The US Food & Drug Administration's (FDA's) approval of subcutaneous immunoglobulin (SCIG) in 2018 provides patients with CIDP more treatment options for maintenance therapy. The different options for administration of IG treatment create the need for information to assist clinicians and patients in choosing the optimal therapeutic approach. Considerations for pharmacokinetics, administration procedures, adverse events, patient variables, and cost will all be discussed in this article.

1. Introduction

Chronic inflammatory demyelinating polyneuropathy (CIDP) is an acquired neurological disorder affecting 1 to 9 per 100,000 adults in the US [1]. Clinically the disorder is characterized by weakness and impaired sensory function that evolves over 2 months or more in a progressive or relapsing fashion. The hallmark electrophysiologic findings of CIDP include conduction velocity slowing, conduction block, temporal dispersion and other evidence of peripheral nerve demyelination [2]. While 89–90% of patients respond to one of the CIDP first-line therapies [3], delayed diagnosis with accumulating axonal damage has the potential to lead to irreversible disability.

First line CIDP treatments with proven efficacy include corticosteroids, plasmapheresis, and immunoglobulin (IG) therapy [4]. While a detailed review of treatment options for CIDP are beyond the scope of this paper and have been discussed elsewhere [5], feasibility and tolerability typically restrict the use of plasma exchange to patients who are severely affected or have treatment-refractory disease. The potential complications of corticosteroids are well known, and while they have

an important treatment role in some patients, chronic corticosteroid exposure is undesirable. While no consensus exists on which first-line intervention is best, these limitations lead many clinicians and patients to opt for immunoglobulin therapy when chronic treatment is needed. IVIG therapy can be administered safely in a hospital setting, an outpatient infusion suite, or at home [6]. Relative to the hospital or office settings, home IVIG administration is geographically and economically advantageous [7] and is thus an increasingly attractive option for patients and insurance carriers. Unlike IVIG which requires administration by a health care professional, subcutaneous immunoglobulin (SCIG) can be self-administered by patients in their home [8]. The flexibility of self-administration may lessen the burden associated with infusions thereby improving a patient's health-related quality of life (HRQoL) [9–12].

In March 2018 SCIG was approved by the FDA for maintenance treatment in adults diagnosed with CIDP [13]. With this approval, individuals with CIDP now have more treatment options for IG maintenance therapy. Treatment decisions for these individuals should be based on conversations between the provider and the patient and

Abbreviations: CIDP, chronic inflammatory demyelinating polyneuropathy; SCIG, subcutaneous immunoglobulin; IVIG, intravenous immunoglobulin

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include the discussion of efficacy and safety as well as patient goals and preferences. The purpose of this review is to describe the clinical and patient considerations when selecting immunoglobulin therapy for CIPD patients.

2. Efficacy and safety of IVIG therapy

IVIG efficacy was established based on five randomized, placebo-controlled trials comparing IVIG to placebo conducted between 1993 and 2008. Based on these trials there is high quality evidence that IVIG is safe and effective for CIPD induction and maintenance treatment [14]. The most informative trial, the Intravenous immune globulin (10% caprylate-chromatography purified) for the treatment of chronic inflammatory demyelinating polyradiculoneuropathy (ICE study) compared IVIG administered as a 2 g/kg loading dose followed by a 1 g/kg every 3 weeks maintenance dose to placebo in 117 patients [15]. The primary outcome measure was the improvement from baseline in disability (measured by the adjusted INCAT disability scale [16]) by 6 weeks and maintained through 24 weeks. At the end of the 24 weeks, responders were enrolled into an extension phase in which they were re-randomized to IVIG or placebo [15].

The percentage of patients with improved disability at 24 weeks was significantly greater in the IVIG group (54%) compared to placebo (21%) and 60% of IVIG responders achieved maximal response by week 6 [15,17]. Significant improvements in MRC muscle strength [18] and grip strength were also documented in the treatment group. In the extension phase, 87% of IVIG-treated patients remained relapse-free over an additional 24 weeks compared to 55% of placebo-treated patients. The primary reason for study discontinuation during the extension phase was CIPD relapse which occurred in placebo-treated patients [17].

Quality of life was assessed in the ICE study by the Short Form-36 (SF-36) [19]. In the first period, patients receiving IVIG reported significant improvements in the physical functioning, role-physical, social functioning, and mental health domains of the SF-36 compared to patients receiving placebo. For those patients who continued receiving IVIG during the extension phase, improvements in SF-36 scores were maintained or improved [20].

Overall, 55% of patients receiving IVIG reported a drug-related adverse event compared to 17% of patients receiving placebo. The most commonly reported IVIG-associated adverse events were headache (32% of patients or 5.2% of infusions), pyrexia (13% of patients or 2.5% of infusions), and hypertension (9% of patients or 1.8% of infusions). Serious adverse events were reported in 5% of patients (0.8% of infusions) receiving IVIG and 8% of patients (1.9% of infusions) receiving placebo. The study discontinuation rate because of adverse events was low: one (2%) patient from each treatment group during the first period, two (4%) patients treated with IVIG during the crossover

period, and one (3%) patient treated with placebo during the extension phase [15].

3. Efficacy and safety of SCIG maintenance therapy

The Polyneuropathy and Treatment with Hizentra (PATH) study [21] evaluated the efficacy and safety of SCIG as maintenance therapy in 172 IVIG-dependent CIPD patients. Participants were randomly assigned to receive SCIG (20% concentration) 0.2 g/kg/wk. (the low dose treatment arm), which is 40% lower than IVIG trial, SCIG 0.4 g/kg/wk. (the high-dose treatment arm), which is 20% higher than IVIG trial, or placebo weekly for 24 weeks. Patients who relapsed during SC therapy were rescued with IVIG and discontinued from the study [21].

The primary outcome was the proportion of patients who relapsed or withdrew from the study for any reason during the 24-week treatment period. Secondary measures included time-to-relapse as well as changes in INCAT disability, grip strength, MRC sum score, and the Inflammatory Neuropathy-Rasch-Built Overall Disability Scale (I-RODS). In the placebo group 63% of patients relapsed or withdrew from the study, compared to 39% in the low-dose group, and 33% in the high-dose group. When only relapse was considered, 56% of patients treated with placebo relapsed in the 24-week period, compared to 33% of low-dose and 19% of high-dose treated patients. There was a statistically significant difference in relapse (with or without inclusion of withdrawal for other reasons) between each treatment group and placebo, but not between the low and high dose groups. Secondary outcome measures analysis showed stable I-RODS disability [22], grip strength, and MRC sum scores in SCIG treated patients at 24 weeks, thus providing additional support that SCIG was superior to placebo to prevent relapse during maintenance therapy [21].

Among patients receiving SCIG, 53% (61 of 115) stated a preference for SCIG treatment compared to 39% (22 of 57) patients in the placebo group. Reasons stated for the preference for SCIG therapy included a gain in independence and fewer side effects. Eighteen percent (21 of 115) of patients receiving SCIG preferred their previous IVIG treatment [21].

During the SC treatment period, treatment-related adverse events were reported in 30% of patients in the low-dose SCIG treatment group, 35% in the high-dose SCIG group, and 18% in the placebo group. The most common adverse events were local infusion site reactions, which were generally mild or moderate. Fifty-four events were reported in 11 (19%) patients receiving low-dose SCIG and 49 events in 17 (29%) patients receiving high-dose SCIG. One serious adverse event (allergic skin reaction) was thought to be SCIG-related [21].

Based on results gathered from the PATH trial, there is now high-quality evidence that SCIG is safe and effective as maintenance treatment for CIPD patients that are dependent on IVIG [21].

There are currently no data available from head-to-head trials

Table 1
Relapse on maintenance therapy: IVIG and SCIG.

IVIG trial (Hughes 2008) [15]				IVIG 1 g/kg every 3 weeks	Placebo	Absolute risk reduction		Hazard ratio	
Relapse during the extension phase ^a (% of patients)				14%	35%	21%		0.19	
Probability of relapse				13%	45%				
^a Defined as worsening of adjusted INCAT disability score by ≥ 1 point from extension phase baseline through week 48.									
SCIG trial (van Schaik 2018) [21]		SCIG	SCIG	Placebo	Absolute risk reduction		Absolute risk reduction	Hazard ratio	hazard ratio
		0.2 g/kg weekly	0.4 g/kg weekly		0.2 g/kg vs placebo	0.4 g/kg vs placebo	0.2 g/kg vs placebo	0.4 g/kg vs placebo	
Primary outcome ^a		39%	33%	63%	25%	30%			
Relapse analysis ^b		33%	19%	56%	23%	37%			
Probability of relapse		35%	22.4%	58.8%			0.48	0.25	

^a Defined as CIPD relapse or were withdrawn from study for any reason.

^b Defined as an increase of ≥ 1 point in the total adjusted INCAT score at any treatment period visit compared with baseline.

Table 2

Risk reduction—PATH findings are similar to those in ICE and Cochran analysis of all IVIG trials in CIDP, suggesting an NNT (number needed to treat) of about 3 [14,15,21,23].

	Relapse ^a		Relapse & Withdrawal	
	PATH	ICE	PATH	ICE
Placebo	56.0	N/A	63.0	42.0
Low dose	33.0	N/A	39.0	N/A
PATH – 0.8 g/kg every 3 weeks				
High dose	19.0	N/A	33.0	13.0
PATH-1.2 g/kg or ICE 1.0 g/kg every 3 weeks				
Relative risk reduction	23%	N/A	24%	N/A
	Low dose		Low dose	
	37%	N/A	30%	29%
	High dose		High dose	

ICE Trial: Defined as worsening of adjusted INCAT disability score by ≥ 1 point from extension phase baseline through week 48.

^a PATH Trial: Defined as an increase of ≥ 1 point in the total adjusted INCAT score at any treatment period visit compared with baseline.

comparing relapse rates in maintenance therapy with IVIG and SCIG, therefore a benefit for one route of administration versus the other has not been determined. Table 1 summarizes the available relapse data for IVIG and SCIG from their respective clinical trials.

From an outcome perspective, patients who were stable on IVIG previously might suffer from an unnecessary increased risk of relapse or withdraw (33.7%) when switched to SCIG (Table 2).

4. Considerations for selection of IG route of administration

With both the IV and SC routes of immunoglobulin administration now available for CIDP maintenance therapy, personalization of treatment has never been more important. While selection of IVIG or SCIG is ultimately a choice best left to the discretion of each patient and their treating physician, several factors warrant consideration so that patients can make informed decisions that best balance their needs, preferences, and lifestyles.

4.1. Pharmacokinetics

After an intravenous infusion of IG the concentration of IgG peaks within minutes [24,25]. Over the subsequent 2–3 days the serum IgG level falls by about half as IgG is distributed into the total extracellular fluid volume [26]. Following this rapid equilibration, IgG is catabolized with first-order kinetics and a half-life of 21–30 days. SCIG is deposited into the subcutaneous tissue and first absorbed into and transported through lymphatics, then enters the bloodstream. Equilibration of the IgG from SCIG into the intravascular space requires about the same amount of time as equilibration of IVIG out of the intravascular compartment, about 2–3 days [24,25].

These pharmacokinetic properties heavily influence the amount and frequency of IVIG or SCIG that is typically administered. With IV administration, because the half-life is 21–30 days, infusions are often repeated every 3 to 4 weeks. Conversely, the amount of immunoglobulin able to be administered into a SC site is less than when administered intravenously and the time to reach a peak serum concentration takes longer. As such, SCIG is typically administered in smaller doses similar to the PATH study, with a frequency of administration usually once or twice per week. As a result, peak serum IgG levels with SC administration are approximately 61% of the peak levels achieved with IVIG [25,27]. However, when considering equivalent doses of SCIG administered weekly or IVIG administered every 21–28 days, trough IgG levels are 10%–20% higher with SCIG administration [28]. The frequent administration of lower doses of IgG

also results in a more stable steady state IgG concentration with SCIG [29]. Immunoglobulin bioavailability may also differ depending on the route of administration. While some studies suggest that equivalent dosing may be effective in CIDP [30], IgG binding to extracellular matrix and IgG degradation in extracellular tissues raises concern that SCIG bioavailability may be 30% lower than IVIG [27,31].

The optimal pharmacokinetic parameters for the treatment of CIDP are unknown. While some have hypothesized that higher trough levels are more important for long term IgG therapy success than higher peak levels [32], this remains unproven. Patients who have clinically meaningful treatment related fluctuations may benefit from the steady state IgG levels of SCIG. The observation that IVIG naïve patients with Guillain-Barré syndrome who had the greatest increments in serum IgG level 2 weeks after treatment also had better 6-month clinical outcomes suggests the importance of high peak serum IgG levels in some patients [33]. In contrast, serum IgG levels trough and peak levels were remarkably constant over time in patients with active but stable CIDP on constant IVIG maintenance treatment, indicating that stable CIDP patients have reached a steady state with a constant distribution rate and turnover of IgG without accumulation over time [32].

While pharmacokinetic parameters are important considerations for optimizing immunoglobulin effectiveness, much remains to be learned regarding how manipulating pharmacokinetics can be advantageous in each individual patient. The key principle is that the dose and treatment intervals required to achieve and maintain any given clinical response vary greatly between individuals. Identification of these differences may allow patients to take advantage of nuanced differences between SCIG and IVIG pharmacokinetics.

4.2. Adverse event profiles

The most common systemic IVIG side effects are headaches and nausea [15]. Many IVIG adverse reactions can be managed by assuring adequate hydration and giving medications such as antihistamines, acetaminophen, and corticosteroids, slowing down the infusion rate, or switching to a different IVIG preparation [34]. Less common but potentially more severe adverse reactions of IVIG include thromboembolic events, including stroke, pulmonary embolism, and myocardial infarction. Advanced age, prolonged immobilization, history of thrombosis or cardiovascular risk factors, and estrogen use, can increase this risk [35]. Aseptic meningitis, renal failure, and pulmonary reactions can also occur. Although rare, anaphylactoid reactions to IVIG has been reported in patients with undetectable IgA and anti-IgA antibodies [36]. Additionally, although not an adverse reaction per se, the need for frequent IV access also becomes limiting for some patients, potentially leading to placement of a port (and all the complications that go along with them).

Systemic side effects of SCIG are similar to IVIG, although the frequency and severity with SCIG infusions are generally less than IVIG [21,30,37]. Especially given that the occurrence of headache in IVIG is more frequent (5.2% of infusions) and potentially precludes the use of IVIG, the observation that SCIG is associated with less frequent and less severe headaches (rate of 0.002 per infusion) is advantageous for some patients [15,21,37]. The slower absorption and lower peak levels of serum IG with SC administration are thought to explain the lower incidence of systemic adverse effects with SCIG [27]. Because SCIG requires no IV access, patients with difficult IV access may be particularly attracted to SCIG. The most common adverse event reported with SCIG, which is unique to SCIG, is local infusion site reactions, such as redness, itching, or swelling. In some patients, local site reactions can be more bothersome and lead to SCIG treatment withdrawal. However, in most patients local site reactions are mild to moderate and improve over time [21].

4.3. Site of care for treatment administration

Currently there are four primary locations where CIDP patients can receive IG therapy: a hospital clinic, physician's office, free-standing infusion clinic, or at home. Due to the need for venous access and monitoring, IVIG therapy has historically been administered in a medical setting such as a hospital, doctors' office, or infusion center. However, with the expansion of home nursing services and the need to reduce healthcare costs, the home setting is rapidly becoming the more common site of IVIG therapy in the US [38,39]. In contrast to IVIG in which a nurse oversees the infusion process, most SCIG maintenance regimens can be administered by the patient or a caregiver without the need for other medical support staff. Patients are permitted to travel with their immunoglobulin, thereby obviating the need to be grounded to an infusion center or home nursing visit at a fixed interval. While these attributes are liberating to some patients, others find the responsibility of preparing and administering SCIG burdensome.

4.4. Dosing and frequency of administration

Based on the ICE trial, EFNS/PNS clinical practice guidelines for the management of CIDP recommend that IVIG be administered as an initial loading dose of 2 g/kg body weight, followed by maintenance infusions of 1 g/kg every 3 weeks [4,15]. The guidelines also acknowledge that, especially for chronic maintenance therapy, there is patient-to-patient variability in the dose and frequency needed for optimal benefit. Each patient should have their dose and frequency individualized to maximize benefit, minimize treatment related fluctuations, and minimize IVIG over exposure. On a practical level this results in most patients receiving IVIG at doses between 0.4 and 1.2 g/kg at frequencies of between 2 and 6 weeks [4].

Based upon results from the PATH study, SCIG doses of 0.2 g/kg and 0.4 g/kg body weight per week are effective for maintenance therapy [21]. These doses reflect a 40% lower dose of IG for the low-dose regimen compared with IVIG dosing and a 20% higher dose when using the high dose SCIG regimen. Although the best approach to transition IVIG to SCIG is unknown, based upon the PATH study and subsequent SCIG package insert the recommended approach is to start at 0.2 g/kg/week (low dose) SCIG. Patients who do not experience satisfactory symptom control can be increased to a higher dose (0.4 g/kg/week) with or without a "rescue" infusion of IVIG before the SCIG dose increase [21,40].

Specific protocols and procedures are recommended when transitioning from IVIG to SCIG. Transition protocols minimize the potential interruption or impact of a change in therapy and ensure that the necessary equipment, supplies, training, and support are available to the patient [41–43]. Considering IgG pharmacokinetic principals, initiation of SCIG is best to follow IVIG by 7–10 days such that the serum IgG concentration remains high enough to smoothly transition to a stable steady state.

4.5. Concentration, volume, and infusion sites

The volume of an IG product administered for an IV or SC infusion will vary based on the concentration of the IG product administered. The typical IVIG product used is a 10% liquid concentration, while a 20% liquid concentration product is used for SCIG [40,44]. The higher concentrated products allow patients to receive smaller volumes of IG thereby requiring less time to infuse [20]. This principal is critical for subcutaneous infusions because only a limited amount of fluid can be absorbed through the subcutaneous tissue. The maximum volume of fluid recommended for infusion through a subcutaneous site is 50 mL [40]. If the recommended dose of SCIG is 0.2 g/kg or 0.4 g/kg, the concentration of SCIG is 20%, and the maximum volume of 50 mL is administered at each infusion site, then a typical 80 kg patient with CIDP requires between 2 and 4 needle sites per week in order to deliver

the necessary volume. While obviating the need for IV access is greatly beneficial to some patients, frequent and multiple subcutaneous needle insertions are undesirable to others [42]. In contrast, IVIG requires only a single needle insertion but can be limited by consistently obtaining satisfactory IV access in some patients [44].

4.6. Infusion rate and duration

Recommended infusion rates for IVIG are product dependent ranging from 2 mg/kg/min up to a maximum of 8 mg/kg/min [44]. For SCIG, the first dose is recommended to be infused at a rate of ≤ 20 mL/h/site but can be quickly increased to 50 mL/h/site as tolerated [40]. The duration of either IVIG or SCIG infusion is dependent on the volume of the product being infused, the number of infusion sites, the infusion rate, and patient tolerance. Considering all these factors, the time to complete an IVIG infusion is 3 to 5 h, while SCIG infusion is 1–1.5 h [15,21]. Recommended IVIG infusion rates for additional tolerance may be much slower [45]. However, since IV is typically infused once every 3 weeks and SC is typically infused weekly, the total cumulative time required to infuse IV or SC is nearly equal over a 3-week period. Table 3 summarizes some of the factors associated with the administration of IVIG and SCIG. Note that these times refer to actual infusion time and do not include preparation and set of equipment.

4.7. Patient preference variables

An analysis of five attributes (mode, frequency, location, # needle sticks, and duration) in preference choice surveys for immunoglobulin routes of administration in primary immune deficiency (PID) patients and caregivers presented several results comparing IV and SC infusions. Location was the most important attribute to patients with the home setting being significantly preferred over a healthcare facility. Frequency, number of needle sticks, and treatment duration were very similarly important to patients, with patients preferring less frequent infusions, fewer needles sticks, and a shorter total monthly length of infusion time. Mode of administration was the least important attribute to patients as well as to caregivers [46]. While the extent to which these observations in PID can be generalized to CIDP is uncertain, what is clear is that administration of immunoglobulin in the home setting is desirable by many (but not all) patients and minimizing the number of needle sticks is intuitively important.

From an autonomy perspective SCIG provides patients with the flexibility to self-administer IG on their preferred schedule. Conversely, handling needles and medical equipment weekly during SCIG infusions without the support of a trained healthcare professional may be limiting to some patients, especially those with upper limb disability [41,42].

Table 3

Attributes of maintenance therapy administration for CIDP: comparison of a 3-week treatment cycle.

Attribute	Three-week treatment cycle ^a		
	SCIG 0.2 g/kg	SCIG 0.4 g/kg	IVIG 1.0 g/kg
	Weekly ^b	Weekly ^c	Every 3 Weeks
Number of infusions/sessions	3	3	1
Number of needle sticks/sites	6	12	1
Total infusion time (total for 3 weeks) ^d	3–4 h	3–4 h	3–5 h
Site of administration	Home or infusion clinic		

^a Based on a person weighing 80 kg.

^b Based on minimum of 2 infusion sites in 1 session per week.

^c Based on a minimum of 4 infusion sites, in 1 session per week.

^d The average times do not include the preparation time required for set-up of each infusion.

While consideration of the infusion setting, number of needle sticks required, duration of infusion, patient autonomy, and patient disability (including overall severity of disability, as well as finger strength and manual dexterity) is necessary for all patients, it is even more important to recognize the uniqueness of each individual patient. Some patients may consider one treatment attribute prohibitive, when another may find that same attribute desirable. Accordingly, presenting each patient with the potential advantages and disadvantages of both routes of administration is prudent.

4.8. Economics

The economics of IV versus SC immunoglobulin therapy for CIDP are not well understood. Two Italian studies analyzed the costs of hospital-based IVIG infusions versus home-based SCIG infusions for patients with autoimmune neuropathies and demonstrated a cost reduction for home-based SCIG compared to hospital-based IVIG [47,48]. However, home administration of IVIG (as opposed to SCIG) was not assessed in these studies. A study conducted in France assessed the cost of home versus hospital-based IVIG treatment in patients with multifocal motor neuropathy and CIDP. The results of this analysis demonstrated that the total annual cost of IVIG therapy was reduced almost in half with the use of home-based administration [7].

Because the immunoglobulin product itself is often the primary expense in the cost of CIDP therapy [39,47,48], any difference in the cost of SCIG and IVIG products bears significant weight on economic comparisons between these two routes of administration. A recent US real-world, retrospective claims database analysis investigated the costs to payers of home-based IV versus SC immunoglobulin in treating PIDD patients. Though not statistically significant, annual mean total PIDD-related costs were nearly \$4000 less for IV than for SC treated patients ($p = .102$). However, annual median total PIDD-related costs were significantly lower for the IVIG group ($p = .002$) [49]. This observation was primarily due to the higher per gram cost of SCIG compared with IVIG in the US [50].

To assess the plausible US economic impact of the two proposed routes of administration for treating CIDP, an analysis using treatment regimens from the ICE and PATH trials for CIDP maintenance therapy were used in a cost calculator to compute the projected costs of treating patients with IVIG and SCIG. This US analysis adopts home administration of IG with doses calculated based on a patient weight of 80 kg. Calculations are set to a 3-week cost period to simplify interpretation of the results. Wholesale acquisition costs per gram for 2019 are used to determine drug costs [50]. Indirect costs such as lost productivity or travel costs were not considered since both IV and SC are both infused in the home setting in the US. The results are described in Table 4.

The results of this hypothetical cost analysis show that the low-dose SCIG regimen (\$8248) results in a \$900 cost savings when compared to the IVIG regimen (\$9131). However, when the high-dose SCIG regimen

is required, the costs to the payer would nearly double (\$16,478 vs \$9131). These results are consistent with the findings from European analyses [47,48], illustrating that the cost of the IG formulation is often the primary driver of the overall cost of CIDP treatment.

5. Discussion

The efficacy and safety of IVIG for induction and maintenance treatment of CIDP is well-established [14]. Completion of the PATH study showing the safety and efficacy of SCIG now provides immunoglobulin-dependent CIDP patients an additional alternative for maintenance therapy [21,30,51]. These new options provide additional considerations for selecting optimal therapy for each individual patient.

Presently there are no data that can predict which patients are “best” candidates for SCIG or IVIG. However, certain characteristics may make one approach more favorable than another in certain clinical scenarios. Difficulty with IV access, intolerable recurrent systemic adverse events such as headache, or patients who desire the autonomy and flexibility of self-scheduled infusions are all factors that may suggest the use of SCIG [37,41,42,52]. Alternatively, patients with an aversion to needles, difficulties with manipulation of the subcutaneous pump or associated supplies, and those with intolerable local site reactions to SCIG are likely to be better candidates for IVIG.

There are many other factors for patients and clinicians to consider that are best done on a case by case basis. Immunoglobulin pharmacokinetics as manifested clinically by the presence of treatment related fluctuations may influence route or frequency of administration in some patients, although it remains to be determined if treatment should aim to eliminate treatment related fluctuations. The duration, frequency, and volume of each individual infusion are important factors for each patient to consider independently as what is a positive for one patient may be a negative for another. The issues surrounding cost are also important. Both on an individual patient and on a societal level, a responsible balance of efficacy, tolerability, and cost is an important goal for which to strive.

For both IVIG and SCIG, justification for ongoing therapy should be based upon objective documentation of examination and other outcome measures that establish immunoglobulin dependency [53,54]. Given the cost and the potential complications of immunoglobulin therapies, patients without unequivocal evidence of benefit should undergo heightened diagnostic scrutiny. Careful monitoring is also needed for IVIG patients undergoing withdrawal or taper trials and for patients that are transitioned from IVIG to SCIG. As some patients may no longer require immunoglobulin therapy, defining relapse as objectively as possible before escalation is important. Likewise, documenting stability or relapse in patients transitioned to SCIG may provide clinicians and patients with the reassurance needed to continue with SCIG or to shift back to IVIG.

The decision to use SCIG or IVIG is one that is best individualized to

Table 4
Three-week cost of maintenance therapy for CIDP: SCIG versus IVIG.

3-Week time period	SCIG		IVIG
Variable	Low-Dose Hizentra (0.2 g/kg/wk)	High-Dose Hizentra (0.4 g/kg/wk)	Gamunex-C 1.0 g/kg/3wk
Product cost/g (WAC)	\$171.5	\$171.5	\$112.4
Dose (g/kg/infusion)	0.2	0.4	1.0
# infusions over 3 weeks	3	3	1.0
Dose over 3 weeks (g/kg)	0.6	1.2	1.0
Total grams/3 weeks (for an 80 kg patient)	48	96	80
IG cost over 3 weeks	\$8232	\$16,464	\$8992
Estimated nursing charge/3 weeks	\$0	\$0	\$139 (Medicare) ^a
Estimated pump charge/3 weeks (Freedom pump)	\$14 ^b	\$14 ^b	N/A
Estimated total cost per 3 weeks	\$8246	\$16,478	\$9131

^a Codes 96,365 and 66 Intravenous infusion (2019 Optum360, LLC. Payers Professional Medicare Part B & DME Fee Schedule).

^b Code E0779 (2019 Optum360, LLC. Payers Professional Medicare Part B & DME Fee Schedule).

each patient's disease characteristics, treatment goals, and lifestyle. There is no "best" method to determine which route of administration is optimal for every patient. Healthcare providers are uniquely positioned to help patients work through these complex issues. Ultimately, the decision is a personal one that ideally maximizes efficacy, tolerability, and quality of life.

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